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Performance evaluation of PV panel surfaces exposed to hydraulic cooling – A review

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ABSTRACT

Hybrid photovoltaic thermal systems consisting of solar panels and solar thermal absorbers are generally used for improving the electrical and thermal efficiency of the photovoltaic panel by cooling. Thus, thermal energy produced can be used for other applications. Depending on solar insolation and location, building integrated photovoltaic (BIPV) systems are designed and developed to fulfill energy needs. This paper reviews the detailed study on the effect of the front surface, back surface and combined front and back surface cooling on the water-based photovoltaic systems. This paper also discusses different designs of solar thermal absorber collectors. The front surface cooling resolves two problems. Firstly, it significantly reduces the PV panel temperature by 22–27 °C depending on the cooling phenomena used and secondly, it cleans the PV panel, thereby increasing the optical efficiency. The PV panel temperature reduction (5–6 °C) in the water spraying cooling system is more than water flowing over the front surface. The photovoltaic thermal absorber collector system is the most economically viable system from the electrical and thermal energy generation perspectives. The evaporative cooling on the backside of PV panels using clay and cotton wick structure could be an effective solution for a standalone PV system. It is observed that dual cooling is suitable for PV panels cooling in a hot arid region but needs further investigation. Finally, the environmental impact and economic feasibility of water-based photovoltaic cooling systems are discussed.

1. Introduction

Industrial development of any country is measured based on available energy. Also, the standard of living of people depends on energy consumption. Energy is needed for various applications such as lighting, operating household appliances, industrial processes, transportation and agriculture applications. Prior to industrial development, fossil fuels were used for power generation. The rising consumption of fossil fuels increased greenhouse gas emissions. According to a report published in 2018 by the International Energy Agency (IEA), a 2.3% global energy increase was observed, leading to an increase in CO_2 production by 1.7% (International Energy Agency, 2018). Looking at population growth, national development and the standard of living, the Indian government has increased the investment in renewable energy. The report published by the National Institution for Transforming India, the energy generation from renewable sources, expects 175 GW by 2022 while solar energy alone (utility-scale, distributed, off-grid/mini-grid) set the target of 100 GW (Aayog, 2015). Solar energy is everlasting, clean and does not damage the environment. The energy supplied by the sun on earth is about 1.8×10^{11} MW, which is many thousand times greater than the energy produced by all sources (Parida et al., 2011).

Solar photovoltaic is a semiconductor material that converts sunlight into electrical energy, reflecting the photovoltaic effect. In recent years, building integrated photovoltaic (BIPV) systems has received significant attention and this technology is considered one of the endorsing non-conventional energy power generation technologies in cities (Eiffert and Kiss, 2000) as a PV panel can be artistically integrated with buildings. The building integrated photovoltaic (BIPV) thermal system developed is based on water type to generate thermal and electrical energy (Hussain et al., 2013; Ji et al., 2006; Kim et al., 2014). Also, the energy and exergy analysis presented is based on building integrated photovoltaic (BIPV) system to increase daily thermal energy gain and electrical performance (Kumar Mishra and Tiwari, 2011; Mishra and Tiwari, 2013; Radziemska, 2009; Yazdanpanahi et al., 2015). The thermal management of PV panels can be done using the active and passive techniques. Active cooling uses the water and air as working medium, whereas passive cooling uses the PCMs, wick structures and heat pipe. The modifications in air channels and sprays nozzles are required for heat transfer enhancement. Also, various types of wick

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Nomenclature

A	area of PV (m^2)
C	heat capacity (J/kg-K)
I	irradiance (W/m^2)
I_c	current (ampere)
I_{sc}	short-circuit current (ampere)
P	power (watt)
T	temperature (°C)
V	voltage (volts)
V_{oc}	open-circuit voltage (volts)

Greek symbols

B	temperature coefficient of photovoltaic cell
η	efficiency
$\dot{m}$	mass flow rate (kg/s)

Subscripts

C	cell
E	electrical
I	inlet/input
m	maximum
o	outlet
oc	open circuit
sc	short circuit

List of abbreviations

AM	air mass
CHP	combined heat and power
ETC	evacuated tube collectors
EWHE	earth water heat exchanger
HDPE	high-density polyethylene
PCM	phase change material
PV	photovoltaic
PVT	photovoltaic thermal
RE	renewable energy

structure, capillary structure, selection of suitable PCMs with modified properties and nanofluids are used for heat transfer enhancement (Maleki et al., 2020). Verma et al. (2020) explained the features and capabilities of different cooling techniques such as heat sink, phase change material, heat pipe, microchannel and thermoelectric generator in thermal management of PV system. It was concluded that a water-based microchannel system is more efficient for hot water generation required for various applications. The PCM-based water system provides a more effective solution for improvement in electrical and thermal efficiency. Siah and Ghadikolaei (2021) reviewed the effect of different hybrid cooling systems on the performance of PV panels and CO_2 reduction in the environment. The increase in cooling water pressure shows improvement in PV panel performance. The hybrid cooling systems such as hybrid PCM, hybrid PCM-water and hybrid PCM-nanofluid techniques effectively reduce CO_2 emission to the atmosphere and shows better enhancement in electrical efficiency compared to the separate use of each system. PCM cooling at 15° inclination angle of PV panel showed the higher CO_2 reduction than the zero-degree inclination.

The solar collector is a device commonly used to produce electrical and thermal energy, depending on applications. Fig. 1 shows classification based on energy conversions such as Solar PV, solar thermal and solar photovoltaic thermal (PVT).

Currently, the conversion of solar radiation into useful work with the photovoltaic system is only 15–20%, while the remaining is converted into heat, which raises the solar panel temperature to 60–80 °C. The increase in cell temperature above the operating limit decreases its electrical conversion efficiency by 0.4–0.65%/°C (Radziemska, 2009). Fig. 2 shows the variation of electrical efficiency over a day with a rise in the PV operating temperature for a typical summer day (Bahaidarah et al., 2016).

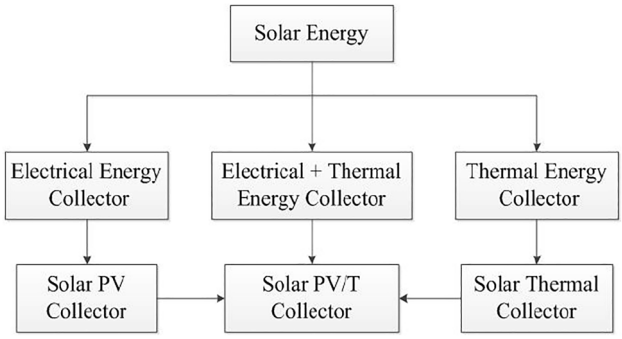


Fig. 1. Solar energy collector classification is based on energy transformation (Satpute and Rajan, 2018).

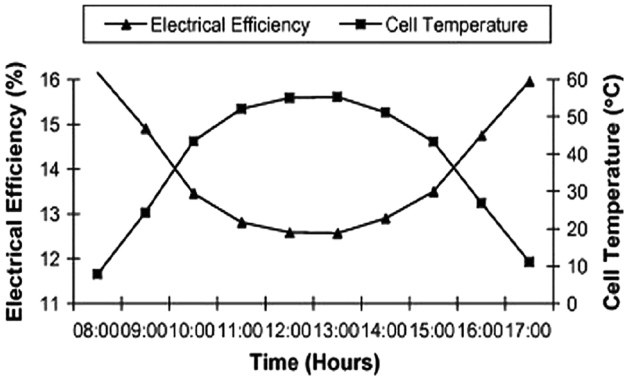


Fig. 2. Effect of temperature rise on electrical efficiency for a typical summer day (Bahaidarah et al., 2016).

2. Temperature regulation techniques

Thermal management techniques play a significant role in controlling the PV module's increase in temperature close to working temperature. Several cooling techniques, such as air cooling, water cooling, heat pipe, microchannel, water immersion, cooling using refrigerant and phase change materials (PCMs) are employed (Sathe and Dhoble, 2017). The PCM packed behind PV panel could maintain a PV cell temperature below the working temperature for some duration and one has to choose PCM according to the environmental condition. Thus, it improves the electrical efficiency and reduces thermal degradation for BIPV systems (Hasan et al., 2015, 2014; Huang, 2011). The use of heat pipe in the photovoltaic thermal system is proposed to investigate the overall conversion efficiency (Gang et al., 2011; Wu et al., 2011). The optimum design of microchannel integrated with a photovoltaic module to maintain its rise in temperature in the hot arid area is presented to improve the electrical output of the solar PV panel (Ali et al., 2015; Reddy et al., 2014). The water immersion technique shows significant improvement in the PV panel electrical performance up to 17.8% in the water depth of 1 cm (Mehrotra et al., 2014). Fig. 3 shows photovoltaic front and back surface cooling techniques.

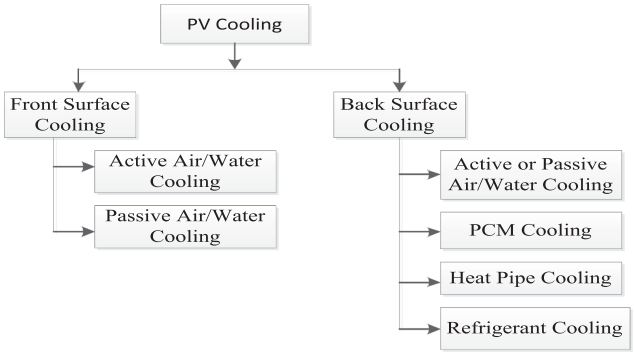


Fig. 3. Photovoltaic front and back surface cooling techniques.

3. Hybrid photovoltaic thermal (PVT) system

The hybrid PVT system is the integration of the photovoltaic module with a PVT system. In the PVT system, the heat extracted from the PV module is used for other applications instead of releasing it into the environment (Zhang et al., 2012). The hybrid PVT system is a simplified approach for increasing PV panel overall efficiency. The PVT systems can be categorized based on air, water, heat pipe, refrigerant and phase change material.

The principal characteristics of the photovoltaic thermal systems are;

a) The system produces both electricity and heat. Heat is useful in other applications.
b) The life cycle cost of a PVT system is higher than the conventional PV system.
c) The electrical and thermal efficiencies of the hybrid PVT systems are much higher than that of the conventional PV system.
d) The PVT sytem reduces thermal degradation and improves the PV module life span.

3.1. Data reduction

The power output, electrical efficiency of the PV module, fill factor (FF), the electrical, thermal and overall efficiency of PVT systems are calculated using equations (Preet, 2018). The fill factor measures the squareness of the IV curve. It shows how a junction is made and how the low series resistance is formed within the cell. As the temperature of the PV cell increases, the fill factor and electrical efficiency decrease, which increases the cost per unit watt.

a) The output power of the PV panel

$$P_m = V_m \times I_m \tag{1}$$

b) The electrical efficiency of the PV panel

$$\eta_{electricalpv} = \frac{P_m}{P_i} = \frac{V_m \times I_m}{I \times A_c} = \frac{V_{oc} \times I_{sc} \times FF}{I \times A_c} \tag{2}$$

c) Fill factor (FF)

$$FF = \frac{P_m}{V_{oc} \times I_{sc}} = \frac{V_m \times I_m}{V_{oc} \times I_{sc}} \tag{3}$$

d) The electrical efficiency of the PVT system

$$\eta_{pv/t} = \frac{V_{oc} \times I_{sc} \times FF \times \beta_c}{I \times A_c} \tag{4}$$

Where β_c is packing factor

$$\beta_c = \frac{\text{area of PV cells}}{\text{area of PV panel}}$$

e) The thermal efficiency of the PVT system

$$\eta_{thermal} = \frac{\dot{m} \times C \times (T_o - T_i)}{I \times A_c} \tag{5}$$

where $\dot{m}$ = mass flow rate (kg/s), C = specific heat (kJ/kgK), T_i = inlet temperature (K), T_o = outlet temperature (K), I = irradiance (W/m^2), A_c = thermal collector area (m^2)

f) The overall efficiency of the PVT system

$$\eta_{overall} = \eta_{electrical} + \eta_{thermal} \tag{6}$$

3.2. Active and passive cooling approach

There are two cooling approaches available to cool the PV panel. First is the active or passive cooling of the PV panels and secondly, the cooling of a photovoltaic thermal system in which both active and passive cooling approaches can be used. Active cooling extract heat from the PV module using an external source of power such as a fan, blower, pump, etc. whereas passive cooling extract heat by natural convection (without using any external source) and discharge it to the environment. Depending on the environmental conditions and design requirements, the cooling method is selected. The use of air and water is widespread among the various cooling techniques to extract heat.

4. Photovoltaic thermal air collectors cooling

Air is freely and easily available working fluid. It can be used for extracting heat from any system. Many investigators have used air as a working fluid for removing heat and improved the PVT system performance. The PVT systems using a flat metal sheet attached to the back surface of an air passage was studied by (Shahsavar and Ameri, 2010; Tonui and Tripanagnostopoulos, 2008). The Numerical model was developed to investigate the effect of various configurations (angles between fins and base plate) of the heat sink on the PV panel performance (Popovici et al., 2016). An enhancement in the PV module's efficiency has been proposed using the aluminium plate heat sink under natural convection and forced convection (El Mays et al., 2017; Soliman et al., 2019). A novel technique of geothermal air cooling system coupled with a PV system for enhancing the PV module performance was developed and studied (Elminshawy et al., 2019).

The comparative analysis was carried out for unglazed, glazed and conventional PVT collectors in the atmospheric condition of Srinagar, India. The results indicated that an unglazed PVT air collector gives improved performance. The unglazed PVT air collector available energy gain is 27% and 29.3% higher than glazed and conventional PVT air collectors (Singh et al., 2012). The investigation of the one pass and two pass PVT systems were analyzed with steady-state conditions. It was found that the two-pass PVT system gives better performance than one pass PVT system. The PV, thermal and PVT efficiencies are 6–7%, 24–28% and 30–35% respectively for one pass, whereas 8–9%, 32–34% and 40–45% respectively for two-pass PVT collector for 1 m length, 200–300 kg/hr mass flow rate and 0.5 pack factor (Abdullah et al., 2018). The PVT air collector was studied for typical summer and winter days to analyze a single pass performance. The investigation shows that electrical, thermal and overall collector efficiencies for summer days are 9%, 22.8% and 47.8%, whereas in winter days are 12.3%, 19.4% and 53.6%, respectively (Amori and Taqi Al-Najjar, 2012).

5. Hydraulic cooling of PV panel

The hydraulic cooling of the PV system gains higher electrical power, electrical efficiency, thermal efficiency and overall efficiency compared with air cooling (Zhu, 2012). Hydraulic cooling is adopted to flow water over the PV module's front or back surface or cool it from both (front and back) sides. Different phenomena are used to cool PV panel, such as direct water flow over the top surface, direct water contact from the backside, nozzle (water spray) arrangement, PVT absorber collector designs, wick structure cooling, or water immersion techniques, etc.

5.1. Front surface cooling of PV panel

This section reviews the experimental and numerical investigations performed by various researchers to enhance photovoltaic panel performance by producing a water film over the front surface. (Krauter, 2004) are the first to investigate the water spraying system over the PV panel's front surface to increase electrical output. A film of water of 1 mm thickness was produced by 12 nozzles placed at the top header, allowing constant water flow at 2 lit/min. The phenomena of producing a thin film of water is shown in Fig. 4. It was observed that the PV panel reflection decreased by 2–3.6%, the operating temperature reduced by 22 °C and the electrical efficiency enhanced by 10%.

(Hosseini et al., 2011) carried out an experimental investigation of the unglazed photovoltaic thermal system and observed a maximum temperature drop of 18.7 °C at a flow rate of 1lit/min. (Moharram et al., 2013) presented a cooling system to minimize the amount of water and electric power required to cool the PV panel. The water was collected at the top headers using a pump, which allowed water to flow at a controlled rate of 29 lit/min through 120 nozzles over the panel's front surface is shown in Fig. 5. The author also proposed a mathematical model (cooling and heating rate model), which tells the starting and stopping of cooling and time required to cool the PV panel. It is reported that the cooling model should be operated at 2 °C /min for 5 min for reducing the PV panel's temperature by 10 °C.

Irwan et al. (2015) have designed and fabricated a solar simulator system to examine the effect of environmental conditions on the PV module in indoor test conditions with and without water cooling. A monocrystalline panel of 50 W was used to study the impact of varying solar irradiance. The water sprayed over the top surface shows a decrease in the surface temperature of 5 °C to 23 °C and an increase in output power by 9–22% for different solar radiation levels. Improving the PV module output performance reduces the system's investment payback period and extends the PV module's life.

Dorobanţu et al. (2013) experimentally investigated the flat-plate PV module's electrical performance and reliability in Romania's outdoor conditions. The film of water produced by a tube having 25 holes of 1.5 mm diameter installed at the top of the panel. The inlet water at 24 °C and the water flow rate of 2 lit/min was maintained The result shows that the front surface cooling reduced the PV panel temperature upto12.5 °C and gained an electrical efficiency of 8.4%.

Smith et al. (2014) carried out an experimental investigation for a low concentrated photovoltaic panel to increase net power output using front surface cooling. The water cooling reduces the panel temperature from 55 °C to 40 °C with a net power output gain of 9.4% over a day. The top surface cooling reduces the deposition of dust, leading to an increase in optical efficiency.

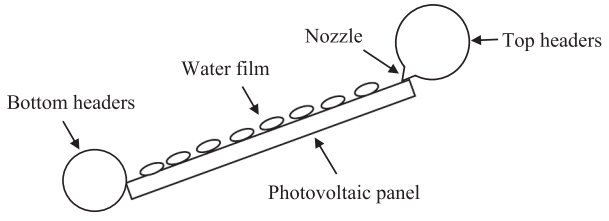


Fig. 4. Schematic diagram of front surface hydraulic cooling (Krauter, 2004).

Matias et al. (2017) performed experiments to investigate the optimum water flow rate required for better power output and efficiency. The tests were conducted with different flow rates of water. The finding shows that at a 0.6 lit/min of an optimum water flow rate, the surface temperature reduces from 70 °C to 50 °C and increases power from 63.09 Wh to 78.74 Wh with an efficiency gain of 22.69%.

Saxena et al. (2018) developed a laboratory-scale experimental setup to examine the effect of continuous and intermittent cooling of the PV panel. The intermittent cooling effect was tested for 3lit/min, 5.3lit/min and 6.2lit/min, while the continuous cooling was performed at 0.6 lit/min. Fig. 6 shows the variation of power output and temperature of the PV module for continuous and intermittent cooling arrangement. Compared to no cooling, an increase in total power output by 18% and 29% was observed for intermittent cooling (for all flow rates) and continuous cooling. A mathematical model validated with experimental results shows that continuous flow requires less water than intermittent flow and surface temperature reduces from 40 °C to 32 °C.

Arefin (2019) studied the PV panel performance by supplying water at different flow rates over the individual PV panel and collector systems front surface is shown in Fig. 7. The experimental and numerical investigation was carried out for 0.5 lit/min to 2 lit/min flow rates with an increment of 0.5 lit/min. The finding shows that at 1.75 lit/min flow rate, the electrical and thermal efficiency is approximately 16% and 64%, respectively and the maximum overall system efficiency obtained is around 80%. The overall system efficiency is five times greater than the PV panel efficiency.

Ramkumar et al. (2016) studied PV system performance using pot water and ordinary water. The water film is produced over the front surface of the panels through the tube slit in it. The result shows that the pot water-cooled PVT panel achieved higher electrical efficiency than ordinary water-cooled PVT panel as the pot water temperature is always 5–8 °C less than the atmospheric temperature. The overall efficiencies of pot water-cooled and ordinary water-cooled PVT panels were 62% and 55% respectively. Pot water cooling could be a better solution for standalone PV panels.

Odeh and Behnia (2009) estimated a long term performance of solar water pumping system using commercial TRNSYS software for pumping water directly from well and from the storage tank. The system consists of a long tube with 32 holes of 5 mm diameter placed at the top, producing a water film over the top surface. The investigation shows a 15% increment in output power in peak hours of sunshine. Table 1 summarizes the investigations carried out by various researchers on PV panel characteristics for front surface cooling using a different cooling approach, mass flow rates and their effects on electrical and thermal efficiency.

5.2. Back surface cooling of PV panel

Another approach adopted for improving the PV panel performance was the extraction of heat from the back surface of PV panels. It was done with different absorber collectors design, spray nozzle arrangement, direct water contact from the backside, or using wick structure.

5.2.1. Photovoltaic thermal absorber collector cooling

The photovoltaic thermal system can be divided into two main types based on the fluids used, i.e., air and water-based. Recently, many investigators have researched air and water-based PVT systems. This section discusses water-based PVT collectors for generating heat and electricity and enhancing electrical and thermal performance using different design configurations of solar thermal absorbers. Sandnes and Rekstad (2002) developed a numerical model and carried out an experimental study on the PVT system with a polymer absorber plate of 0.82 m in length and 0.59 m in width attached to the backside of the panel and fluid flow in a square channel that contains granules of

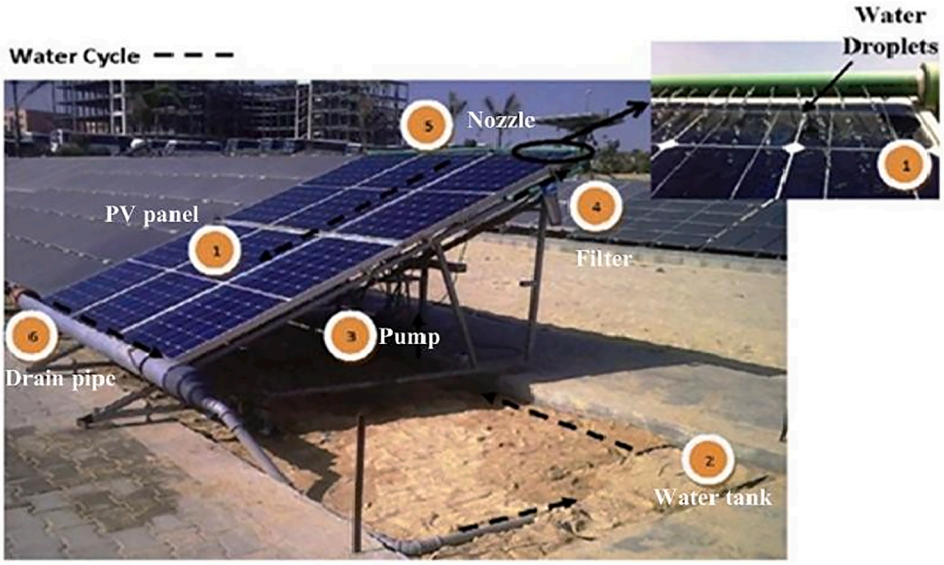


Fig. 5. Photograph of the experimental setup of the PVT system (Moharram et al., 2013).

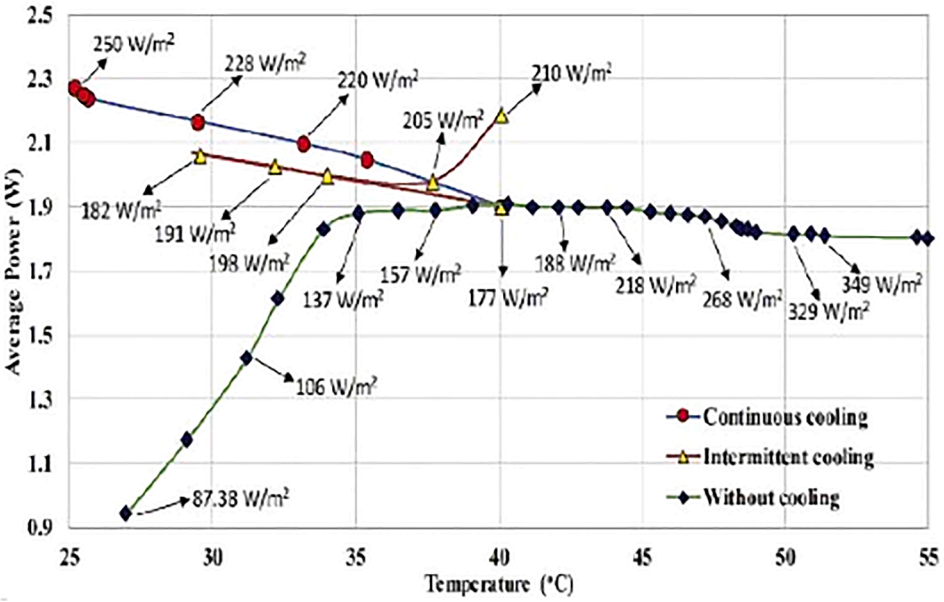


Fig. 6. Variation of power output and temperature of the PV module for a different cooling arrangement (Saxena et al., 2018).

ceramics is shown in Fig. 8. The ceramic granules extract the heat from the PV panel and water passing through a channel absorbed the heat from granules. The result shows that the polymer absorber maintains a temperature below 45 °C.

Ibrahim et al. (2009) have designed, investigated and analyzed PVT performance for various operating conditions such as solar irradiance, ambient temperature and mass flow rates. Seven new types of absorber collectors in the form of rectangular and round tubes were designed. The simulations for all configurations were carried out to find the best geometry. The simulation result shows that spiral flow design is the best absorber design at 0.01 kg/s flow rate. This design shows an enhancement in electrical and thermal efficiency by 11.98% and 50.12%.

The PVT system performance was investigated experimentally using a fully wetted absorber collector by Kim and Kim (2012). The PVT collector shows a 15% increment in electrical efficiency and a 65% increase in overall thermal output. Alzaabi et al. (2014) experimentally examined the hybrid photovoltaic thermal system performance by attaching the rectangular copper tubes to the PV module's backside. A 5 W pump circulates water through these tubes at a flow rate of 5.4 lit/min is shown in Fig. 9. Due to water circulation, a significant decrease in surface temperature by 15–20% is observed and, consequently, an increase in electric output power by 15–20%.

PVT system performance was investigated using active and passive cooling approaches by Ozgoren et al. (2013). The result shows that active cooling reduces operating temperature from 65 °C to 32 °C and electrical conversion efficiency increases to 13.6% compared to passive cooling. Rawat et al. (2014) experimentally examine the performance of the sheet and tube type collectors. The experimental setup consists of a copper sheet attached to the PV module's backside, which acts as an absorber. The copper tubes were attached to the copper sheet. The water was circulated through these tubes at 0.002 kg/s using the 18 W pump. The result shows that the proposed system increases electrical efficiency from 7.58% to 8.27% and thermal efficiency obtained around 50.10%.

Xu et al. (2015) carried out simultaneously the experimental

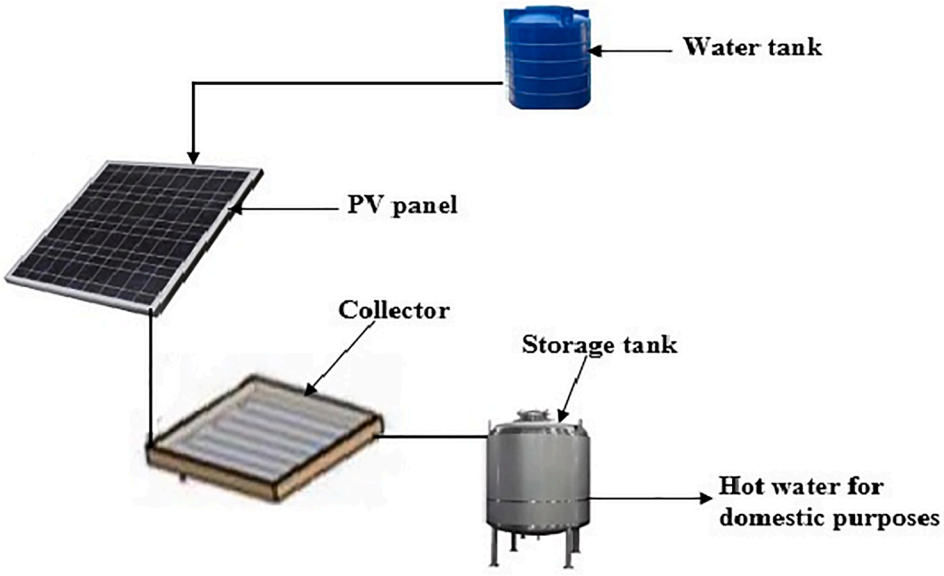


Fig. 7. Front surface cooling water-based PVT system (Arefin, 2019).

Table 1
Literature review summary for front surface cooling of the PV panel.

Authors	Characteristics of PV panel	Cooling approach	Mass flow rates	Electrical Efficiency	Thermal Efficiency	Efficiency gain	Input water temperature	Energy used for operating the cooling system
(Krauter, 2004)	Type: multi-crystalline Power: 53 W Area: 0.286 m^2	Active	2 lit/min	10.3%	–	8–9%	–	–
(Moharram et al., 2013)	Type: Monocrystalline Power: 185 W Area: 1.258 m^2	Active	29 lit/min	12.5%	–	–	35 °C and 25 °C	–
(Matias et al., 2017)	Power: 140 W	Active	1 lit/min –0.125 lit/min	–	–	22.69% (at 0.6 lit/min)	–	~1.33 Wh
(Kordzadeh, 2010)	Type: Crystalline Power: 90 W & 135 W (Array)	Active	100–1100 lit/hr	–	–	0.35–0.68%	–	–
(Smith et al., 2014)	Type: Monocrystalline Power: 175 W	Active	7 lit/min	–	–	–	–	~56Wh
(Odeh and Behnia, 2009)	Type: Multi crystal Power: 60 W	Active	4 lit/min	–	–	4–10%	25 °C	–
(Arefin, 2019)	Power: 20 W Area: 0.22 m^2	Active	0.5, 1, 1.25, 1.5, 1.75 and 2 lit/min	16%	64%	0.8–1.5%	27–33 °C (panel inlet) and 28–36 °C (collector inlet)	~1.5Wh
(Saxena et al., 2018)	Power: 40 W Area: 0.24 m^2	Passive	3, 5.3 and 6.2 lit/min	–	–	–	–	Nil
(Ramkumar et al., 2016)	Type: Monocrystalline Power: 40 W Area: 0.315 m^2	Passive	0.3 kg/hr	8.6–9.4% (ordinary water) and 8.4% – 9.8% (pot water)	46–53%	5.68%	35 °C (ordinary water) and 32 °C (pot water)	Nil

investigations in the laboratory as well as in the field on the PV panel and PVT system. The laboratory assessment was carried out using a single PV panel and a single PVT system, whereas field performance was carried out using 15 standalone PV panels and 15 PVT panels. The PVT system consists of a thermal absorber (flat metal sheet) attached to the PV panel's backside. Fig. 10 shows the daily electricity output generation for PV and PVT systems. The laboratory experimentation of the PVT system shows a higher electrical efficiency (16.8%) than the standalone PV panel (11.8%) and the corresponding thermal efficiency of 65%.

The PVT performance of mono and polycrystalline panel was compared for the same power output by Bashir et al. (2018). The experimental investigation shows an average temperature decrease of 13.6% and 7% for mono and polycrystalline modules at 0.02 kg/sec flow rate. He et al. (2011) developed a PVT system based on the natural circulation of water to improve the system's energy-saving efficiency. The primary-energy saving efficiency of the developed system was found to be 60–75% much higher than the conventional system.

The cold-water circulation enhances the heat extraction rate and improved PV performance compared to ordinary water. Hussien et al. (2015) studied the effect of cold water circulation through copper tubes for 0.1 lit/sec to 0.3 lit/sec. The higher flow rate of 0.3 lit/sec, reduces the panel temperature to 70.1 °C and increases the panel's electrical

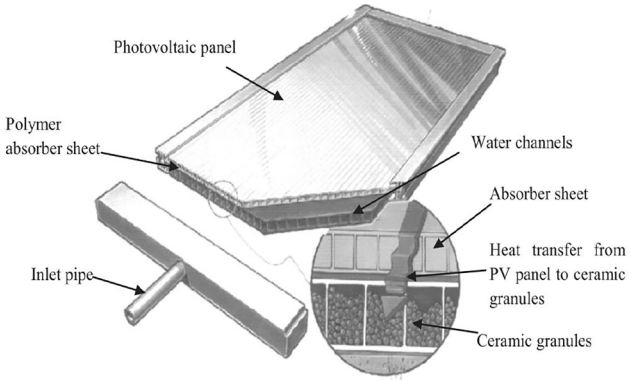


Fig. 8. Polyphenylene oxide plastic absorber is filled with ceramic granules (Sandnes and Rekstad, 2002).

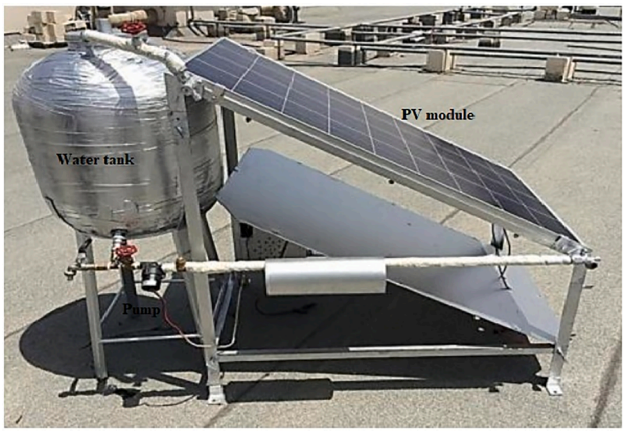


Fig. 9. Hybrid photovoltaic thermal system (Alzaabi et al., 2014).

performance compared to 0.1 lit/sec (76.8 °C) and 0.2 lit/sec (74.5 °C).

Mohsenzadeh and Hosseini (2015) observed that heat absorbed by the PV module was not sufficient to produce hot water. The reflectors were used to increase the solar intensity and to extract heat more efficiently. The two steel plate reflectors of 820 mm × 535 mm in size were used to concentrate solar insolation. The water was circulated from the top header through 12 holes at different water flow rates of 0.012 kg/s, 0.025 kg/s and 0.05 kg/s. Fig. 11 shows a schematic diagram of the PVT collector and ETC collector for evaluating electrical and thermal performance. Due to water circulation, the temperature of PV module was reduced by 8.2 °C compared to the reference module and during experimentation, the panel temperature was maintained below 50 °C.

Kazem (2019) have analyzed and evaluated the PVT panel performance for a direct flow absorber collector. The experiment was conducted on a 100 W PV panel at a flow rate of 0.02 kg/s. It is observed that the maximum electrical power obtained (67 W) for the PVT system is 6% higher than the reference PV system. Rejeb et al. (2015) have numerically investigated the dynamic electrical and thermal behavior of sheet and tube type PVT collectors. A mathematical model was developed to optimize the performance of the PVT collector based on energy balance for six components is shown in Fig. 12. The programmed simulation was solved in FORTRAN 90 software to obtain the best results. The numerical investigation shows that the increase in packing factor and coefficient of heat conduction between PV panel and the absorber plate leads to an increase in electrical output, while both reduce with the increase in water inlet temperature and the solar irradiance. The copper metal was considered for the evaluation of heat transfer coefficient by conduction and the thickness of copper tube is 0.0012 m. The thermal efficiency of PVT water collector increases with the coefficient of heat conduction and solar irradiance between the panel and the absorber plate. However, it reduces with the increasing temperature of the inlet water and the packing factor. Also, if the heat conduction coefficient between the PV panel and absorber plate increases beyond 300 W/m^{2}k, it does not significantly enhance the electrical as well as thermal output.

Bahaidarah et al. (2013a) have numerically investigated PVT performance at a 3.6 lit/min flow rate. The results show that active cooling reduces the surface temperature by 19.3%, leading to an increase in electrical output by 9%. Nahar et al. (2017) presented a 3D numerical study for a new pancake shape thermal collector. This flow channel design reduces the cell temperature by 37%.

Misha et al. (2019) have evaluated the PVT performance using a dual oscillating absorber is shown in Fig. 13 for Malaysian climatic conditions. The numerical simulation was carried out for different mass flow

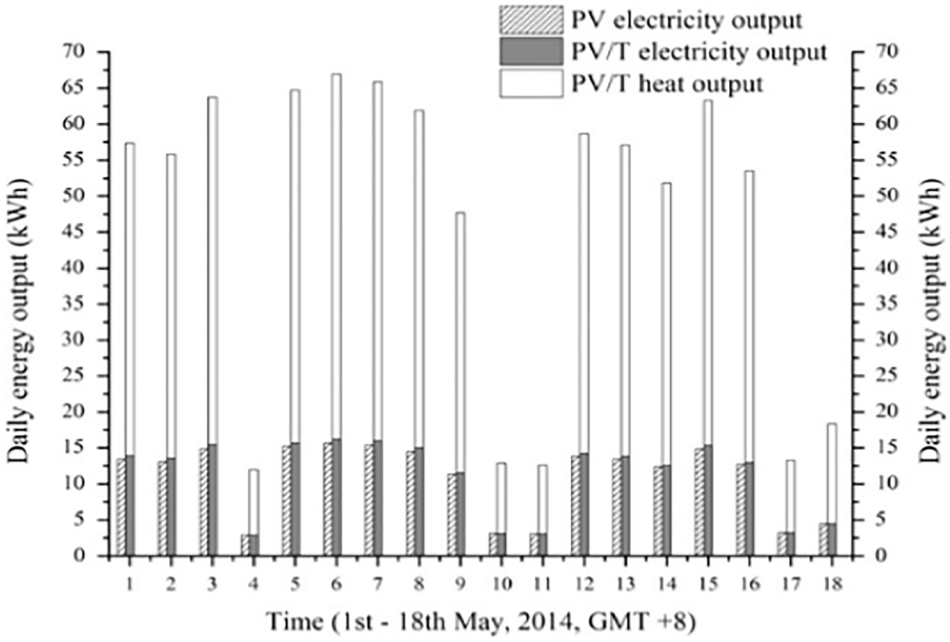


Fig. 10. Daily electricity output for PV and PVT systems (Xu et al., 2015).

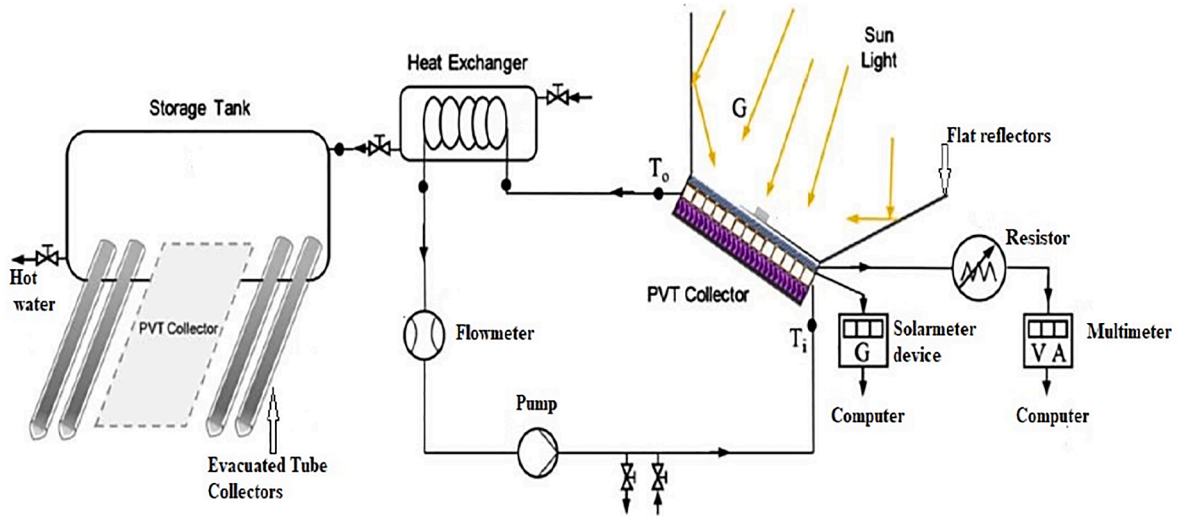


Fig. 11. Schematic diagram of PVT collector and ETC collector for evaluation of electrical and thermal performance (Mohsenzadeh and Hosseini, 2015).

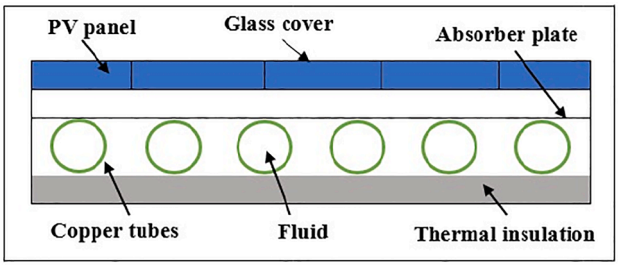


Fig. 12. Photovoltaic thermal (PVT) collector water (Rejeb et al., 2015).

rates and varying solar radiation levels in Ansys 19.2 software. The study shows that the numerical findings were in good agreement with the results of the experiment. The results show that the electric power and electrical efficiencies are 70.60 W and 75.13 W respectively for the PV system and 10.86% and 11.71% for a PVT system, at a 6 lit/min flow rate. With an increase in the mass flow rate, the electrical performance increases and decreases with the increase in solar radiation level.

Buonomano et al. (2016) performed experiments throughout the year and carried out numerical simulations to compare the performance of PV and PVT collectors and find the economic feasibility of the system. The result shows that the maximum temperature drop of 10 °C was observed. The average electrical efficiency for both PV and PVT systems is nearly the same, i.e., 15%, while the PVT collector's average thermal efficiency obtains around 13%. Alobaid et al. (2018) studied the effect of the PVT system on water outlet and solar cell temperature. Also, a system's thermal and electrical performance was mathematically analyzed

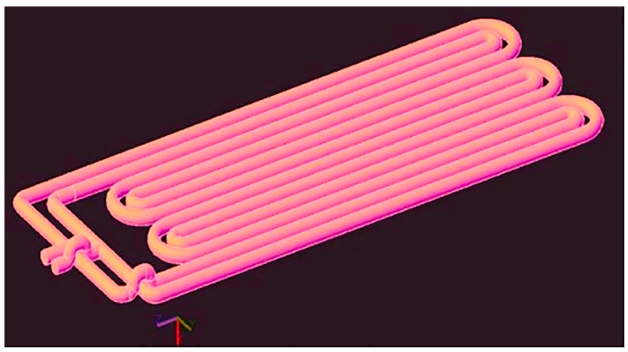


Fig. 13. Design of dual oscillating absorber (Misha et al., 2019).

for different water inlet temperature inputs, which was experimentally evaluated by Bahaidarah et al. (2013b). The numerical results are in good agreement with the experimental results. It is observed that with an increase in inlet water temperature, electrical and thermal efficiency decreased. At lower inlet water temperature, the maximum electrical and thermal efficiency obtained was 13.7% and 65%, respectively.

Jakhar et al. (2016) developed an unglazed PVT system based on earth water heat exchanger (EWHE) to improve PV performance by varying operating parameters. Fig. 14 shows a schematic diagram of the EWHE based unglazed PVT system. The system is designed and simulated using TRNSYS software for different mass flow rates, pipe length, pipe material and buried pipe diameter. The result shows that at 0.018 kg/sec flow rate, the peak cell temperature reduces from 79.31 °C to 47.13 °C and increases power output by 23.16 W using EWHE cooling. Also, for a pipe length of 10–50 m, the maximum surface temperature drops from 60 °C to 42.89 °C. It is also suggested that the HDPE pipe diameter of 12 mm can be used for practical application.

Anand and Srinivas (2017) developed a mathematical model for assessing the PVT desalination system's performance based on the humidification and dehumidification process. The system performance was assessed in terms of efficiency and improvement in desalination by varying solar intensity and water flow rates through an absorber tube. The flow rate was varied from 30 to 110 kg/hr. The finding shows that at 110 kg/hr water flow rate and solar intensity of 800 W/m^2, the maximum PV panel efficiency obtained 16.59%.

Hussein et al. (2017) developed a new design of the PVT/nanofluids system to study the enhancement in electrical and thermal performance. It consists of a heat exchanger in which water circulating pipes were attached at the panel's backside. A copper tube of 11 mm inner diameter and 1 mm thickness with the surface plate of 2 mm thickness was used to increase thermal conductivity (efficiency). The analysis was carried out at different concentration ratios of nanofluid (Zn-H_2O) varying from 0.1 to 0.5% and at 1, 1.5 and 2 lit/min flow rates. The experimental result shows that water cooling at a 2 lit/min flow rate reduces the panel's temperature from 76 °C to 70 °C. With the nanofluid addition at an optimum concentration of 0.3%, the panel temperature was reduced to 58 °C.

Ghadiri et al. (2015) performed experiments to study the combined effect of ferrofluid (Fe_3O_4/water) and magnetic field on the PVT system performance. Fig. 15 shows the schematic diagram of the magnetic field-based PVT system. The enhancement in the overall performance of the PV panel was compared with and without applying a magnetic field. The ferrofluid at 1% and 3% by weight concentration was passed through an absorber collector at a 30 lit/hr flow rate. The investigation shows that

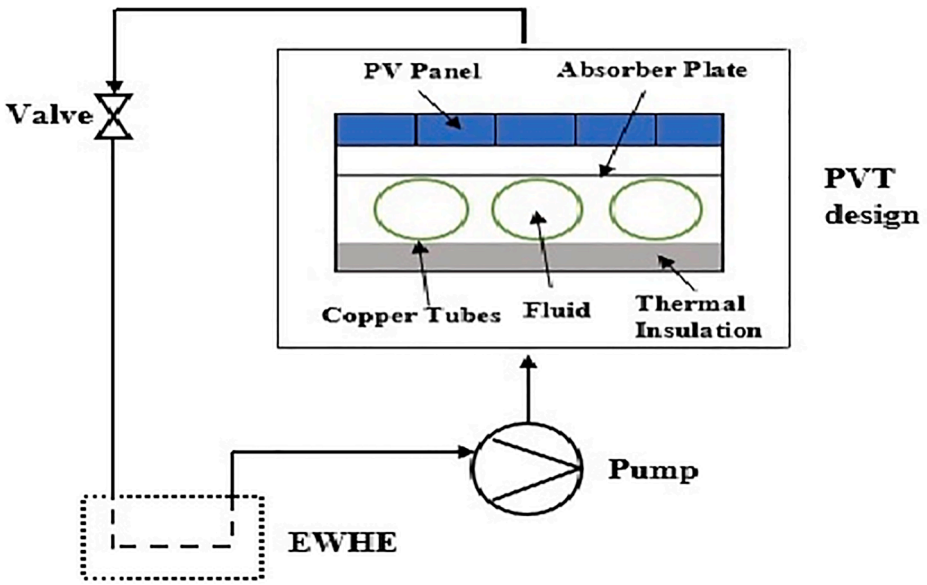


Fig. 14. Schematic diagram of the unglazed PVT system (Jakhar et al., 2016).

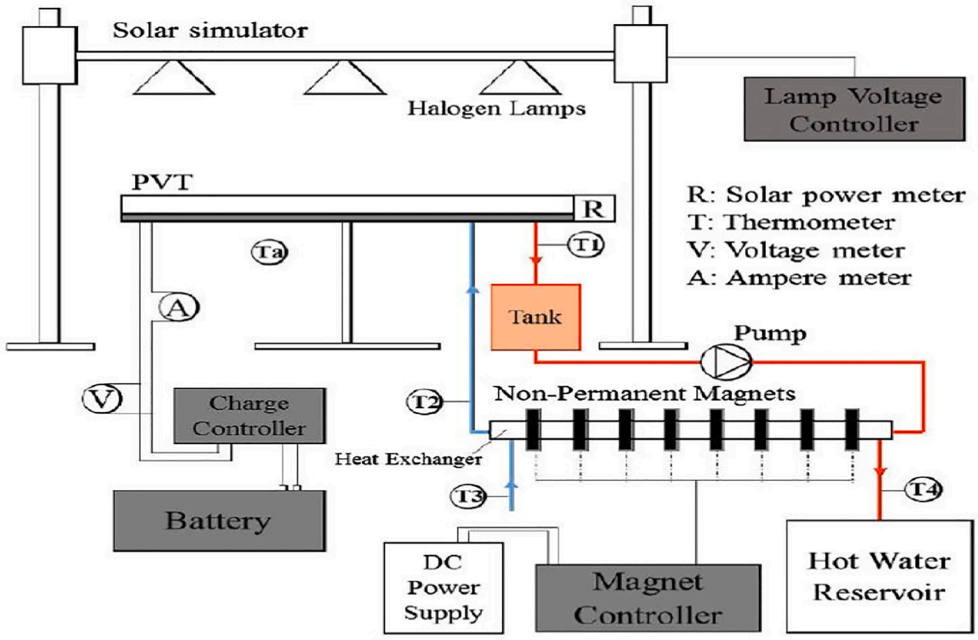


Fig. 15. Schematic diagram of the magnetic field based PVT system (Ghadiri et al., 2015).

3% of weight concentration increases overall efficiency by 76% without a magnetic field. Also, it is observed that a magnetic field of 50 Hz increases the efficiency to 79%.

Sardarabadi and Passandideh-Fard (2016) presented a numerical and experimental investigation of sheet and tube type absorber water-based nanofluids PVT system. The electrical and thermal performance was investigated using three different nanofluids (Al_2O_3, TiO_2 and ZnO) with base fluid as deionized water (0.2% by weight). The numerical and experimental result shows that the electrical performance of water/TiO_2 (6.54%) and water/ZnO (6.46%) nanofluids are more than those of the water/Al_2O_3 (6.36%) nanofluid and deionized water (5.48%). Whereas, the thermal performance of water/ZnO nanofluid is higher than that of the other two nanofluid and deionized water. With the increasing percentage of mass fractions of nanoparticles up to 10%, the system's thermal performance increased by 4 times. However, the surface temperature decreased by only 2%. It was also observed that a high mass fraction of nanoparticles considerably improved the system's thermal performance. However, there is no significant improvement in electrical efficiency.

Mustafa et al. (2017) have evaluated the temperature distribution of each layer of the PV panel and studied its effects on electrical and thermal performance. The numerical investigation was performed using TiO_2 nanofluids for various solid volume fractions, different mass flow rates and solar irradiance levels. Their study shows that a lower volume concentration, higher mass flow rate and lower solar irradiance demonstrate improved performance. The electrical and thermal efficiencies of PVT/nanofluids obtained around 14.38% and 71.71%, respectively, at 0.0174 kg/sec flow rate and 650 W/m^2 solar irradiance.

Nasrin et al. (2019) developed a 3D numerical model for a low concentrated Fresnel lens PVT/Nanofluids system with a baffle based

solar thermal collector is shown in Fig. 16. The simulation was performed using COSMOL Multiphysics software for different nanofluids with various solid concentrations and varied inlet temperature from 20 °C to 40 °C. The findings showed that the best outcomes were obtained for Ag/H_2O nanofluid in terms of electrical, thermal and overall efficiency at optimum solid concentrations of 2% and decreased the cell temperature by 2.35 °C. The increase in thermal efficiencies of the PVT system operated by the solid concentration of 2% of nanofluids such as Ag/H_2O, Cu/H_2O and Al/H_2O were 7.49%, 7.08% and 4.97%, respectively, compared with water.

In addition, the performance of various designs of PVT/nanofluids water-based collectors at various solid concentrations, different mass flow rates and outdoor operating conditions was evaluated numerically and experimentally (Binti Rukman et al., 2019; Chamkha and Selimefendigil, 2018; Lee et al., 2019). Table 2 summarizes the investigations carried out by various researchers on PV panel's characteristics for back surface cooling with different solar thermal absorber collectors using a different cooling approach, mass flow rates and their effects on electrical and thermal efficiency.

5.2.2. Evaporative cooling of PV panel

This section describes the back surface evaporative cooling of the PV panels using clay, cotton wick structure and cotton wick structure with heat spreaders. Alami (2014) proposed an evaporating cooling system in which synthetic clay layers of three different thicknesses of 2 mm, 4 mm and 6 mm each weighing 1 kg, 1.5 kg and 2 kg respectively were used. One side of clay was covered with a copper sheet while the other side was exposed to the backside of the PV panel, which cools it by evaporating water in clay. It was found that 2 mm thick clay shows enhancement in the evaporation rate compared to other clay thickness. The proposed system shows an improvement in output voltage by 19.4% and power by 19.1% compared to the PV system without clay.

Passive cooling with water-based cotton wick structure nanofluids such as CuO/water and Al_2O_3/water were experimentally investigated to improve PV panel performance. A wick structure of 7 mm in diameter was circularly placed on the backside of the panel. The experimental results show that when the cotton wick is soaked in water, the PV panel temperature drops to 54 °C. While it is soaked in nanofluid, the PV panel temperature reduces to 59 °C. It was observed that the cooling system performance deteriorated with nanofluid compared to water due to the adhere of nanoparticles on the wick fibers that affect wick structures capillary action (Chandrasekar et al., 2013). The phenomena of producing evaporative cooling using cotton wick structures with a heat spreader is shown in Fig. 17. The experimental investigation shows that panels temperature reduces from 49.2 °C to 43.3 °C, leading to improved electrical efficiency by 14% (Chandrasekar and Senthilkumar, 2015). The PV panel's electrical characteristics and efficiency for copper heat sink with wick structures are higher than the aluminium heat sink (Parkunam et al., 2019).

Haidar et al. (2018) fabricated an experimental setup based on evaporative cooling to bring the panel temperature to a lower value. The temperature drop obtained with evaporative cooling was more than 20 °C. It was also observed that evaporative cooling improves the power plant availability and reliability when coupled with the generation system and operated at peak hours.

5.3 Front and back surface cooling of the PV panel

This section describes the effect of the front and back surface cooling using a spray nozzle system, direct water contact system and producing water film over the surface. Nižetić et al. (2016) performed experiments to study water spray's influence on the front surface, a back surface and combined front and back surface of the PV panel to cool it during peak solar irradiance. The experimental setup consists of 10 nozzles on the front side and 10 nozzles to the backside of panel. A plastic tube of 4 mm diameter was used to supply water. The water flow rate through the nozzle varied from 44 lit/hr to 225 lit/hr. Fig. 18 shows the variation of PV operating temperature for various cooling regimes during the peak solar irradiance. The result shows that due to water spray, the average surface temperature reduces from 54 °C to 24 °C, which increases productive power output and electrical efficiency by 7.7% and 15.92%, respectively.

Hachicha et al. (2015) have experimentally investigated the PV system performance by dual cooling in a hot arid region. Simple flow under gravity was used to cool the front surface and direct contact water cooling from the backside. The author reported that the front surface cooling was more effective than back surface cooling. Fig. 19 shows the effect of the front, back and dual surface cooling on the module's temperature and power output. The front surface cooling reduces the surface temperature by 11%, while back surface cooling by 9% and dual cooling reduces surface temperature by 18%.

Ahmad et al. (2018) suggested a new technique for cooling the PV module's front and back surface to mitigate the effect of high-temperature in Qatar's environmental conditions. The system consists of a 4 mm aluminium sheet attached to the backside which, holds water directly in contact with the panel. An electronic valve operated by the microcontroller regulates the flow of water over the front surface through the PVC pipe installed at the top edge. The result shows that dual cooling reduces the temperature of the panel by around 30 °C and the average power output is increased by 5.6%. The separate cooling for the front and the back surface increases power output by 4% and 5.4%, respectively. Table 3 summarizes the investigations carried out by

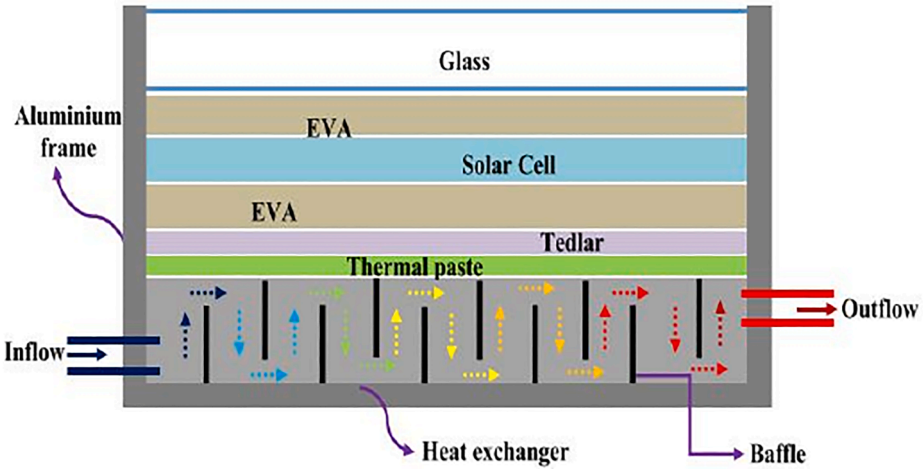


Fig. 16. Water PVT/Nanofluids collector system (Nasrin et al., 2019).

Table 2
Literature summary for back surface cooling using various PVT absorber designs.

Authors	Characteristics of PV panel	Absorber type and flow patterns	Cooling approach	Mass flow rate	Electrical Efficiency	Thermal Efficiency	Efficiency gain	Input water temperature	Energy used for operating the cooling system
(Sandnes and Rekstad, 2002)	Type: Monocrystalline Power: 36 W Area: 0.32 m^2	Polymer thermal absorber and square channels	Active	–	–	–	–	10–12 °C	–
(Ibrahim et al., 2009)	Type: Mono and Polycrystalline	spiral flow design and rectangular hollow tubes of Stainless steel	Active	0.01 kg/s	11.98%	50.12%	–	25 °C	–
(Rawat et al., 2014)	Type: Polycrystalline Power:37 W Area: 0.3216 m^2	Copper sheet and tube type absorber	Active	0.002 kg/s	7.57%	50.1%	0.16–0.08%	23 °C	~58Wh
(Kim and Kim, 2012)	Type: Monocrystalline Power: 240 W Area: 1.65 m^2	Sheet and tube type absorber	Active	0.02 kg/s	15%	50%	2%	10–40 °C	–
(Bashir et al., 2018)	Type: Mono and Polycrystalline Power: 40 W Area: 0.29 and 0.26 m^2	Duct type of solar collector	Passive	0.02 kg/s	12.7%	–	1.5% (monocrystalline) & 0.75% (polycrystalline)	–	Nil
(He et al., 2011)	Type: Monocrystalline Area: 0.014 m^2	Sheet and tube type absorber	Passive	–	10%	40%	–	–	Nil
(Aste et al., 2015)	Type: Amorphous Power: 125 W	Roll-bond flat plate type absorber	Active	0.066 kg/s	13.2%	28.8%	–	10 °C	~8.5 Wh
(Connor et al., 2018)	Type: PV array (72 panels) Power: 250 W Area: 1.72 m^2	sheet and tube type absorber (Copper, Aluminium, steel and polymer)	Active	0.1 kg/s	13.7%	72–83%	1.2%	14 °C	–
(Hussien et al., 2015)	Power: 65 W Area: 0.654 m^2	Copper sheet and tube type absorber	Active	0.1–0.3 lit/sec	9.6%	12.3%	1.8% (at 0.2 lit/sec)	20 °C	–
(Bahaidarah et al., 2013a)	Type: Monocrystalline Power: 230 W Area: 1.24 m^2	Solar thermal collector	Active	3.6 lit/min	9%	–	3%	18 °C	~130Wh
(Kim et al., 2014)	Type: Monocrystalline Power:240 W Area: 1.65 m^2	Aluminium sheet and tube type absorber	Active	10 lit/min	17%	30%	1%	25 °C	–
(Mohsenzadeh and Hosseini, 2015)	Type: Monocrystalline Power: 60 W	Brass channels thermal absorber	Active	–	11.9%	62.98%	1.1%	25 °C	~115Wh
(Buonomano et al., 2016)	Type: Polycrystalline Power: 250 W Area:6.5 m^2	Janus PVT collector	Active	0.5–20 lit/min	11.6%	14.4%	–	27 °C	~61Wh
(Alzaabi et al., 2014)	Type: Polycrystalline Power:100 W Area: 0.784 m^2	Parallel-tube Flat-plate collector	Active	5.4 lit/min	11.93%	60% to 70%	–	35.5 °C, 38.0 °C and 39.2 °C	~35Wh
(Xu et al., 2015)	Type: PV array Power: 210 W Area: 1.28 m^2	Super thin conductive thermal absorber	Active	100 lit/hr	16.8%	65%	3.5%	30 °C	–
(Ghadiri et al., 2015)	Type: Monocrystalline Power: 40 W	Copper sheet and tube type absorber	Active	30 lit/hr	7.28%	72.59%	4.8% (3 wt% ferrofluid)	–	–
(Li et al., 2014)	Type: Semitransparent Area: 0.112 m^2	square tube collector	Active	–	11.5%	39.5%	0.113–0.095%	19.4–44.5 °C	~11Wh
(Misha et al., 2019)	Type: Monocrystalline (semi-flexible) Power: 100 W Area: 0.6372 m^2	Dual oscillating absorber collector	Active	2, 4, 5 and 6 lit/min	11.71% (at 6 lit/min)	59.6%	1.49%	26 °C	~126Wh
(Hussein et al., 2017)	Type: Monocrystalline Power: 110 W	Serpentine flow collector and copper sheet pipe	Active	2 lit/min	6.5–7.8% (Zn-H_2O)	60%	1.3%	–	~72Wh

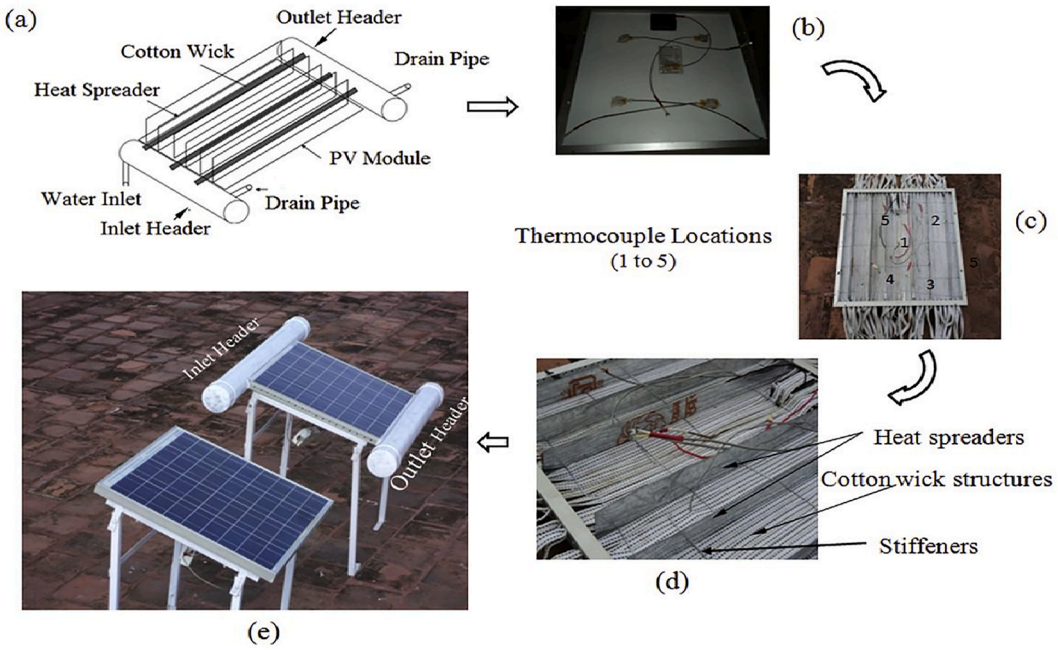


Fig. 17. Fabricated experimental setup with cotton wick structure (Chandrasekar and Senthilkumar, 2015).

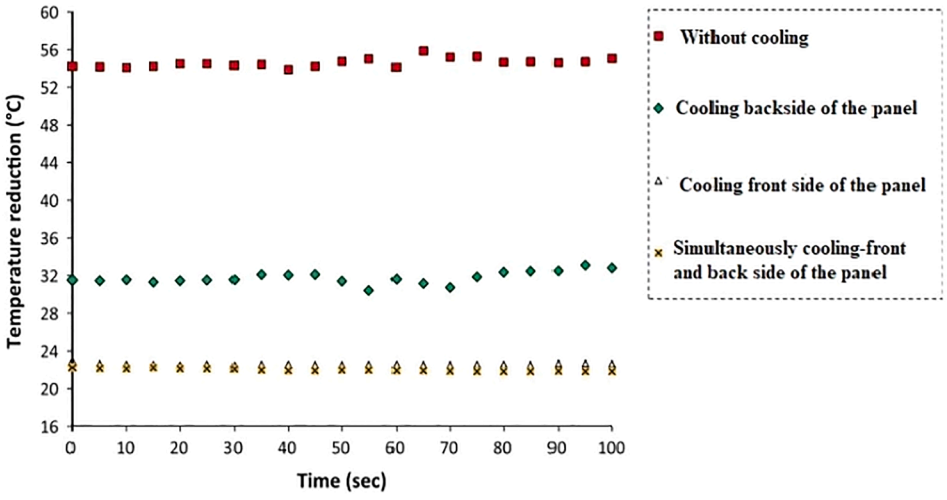


Fig. 18. Variation of PV operating temperature for various cooling arrangements during the peak solar irradiance (Nižetić et al., 2016).

various researchers on PV panel characteristics for the front and back surface with the different cooling approaches, mass flow rates and their effect on the PV panel performance.

Table 4 summarizes the investigations carried out by various researchers for the front, a back and combined front and back surfaces with different cooling phenomena used and their effect on the PV panel performance.

6. Water losses in a photovoltaic cooling system

In a photovoltaic cooling system, the water losses are mainly due to the evaporation and leakage from the PV panel. Moharram et al. (2013) observed that the water loss due to evaporation per month is about 5% of the total water in the water spraying system. Thus, this system is favourable for PV station installation in deserts. Smith et al. (2014) observed that the water losses due to the continuous water flow over the front surface are small compared to the water spraying. Saxena et al. (2018) reported that if the cooling of PV starts beyond 40 °C, the water loss due to evaporation increases in continuous water flow over the front surface. When the water flows over the front surface, the loss of water due to evaporation and leakage was observed by Arefin (2019). This loss of water is not constant for all days, but it varies with the solar intensity and temperature of the PV panel. At a lower mass flow rate, the evaporation rate is higher; however, the evaporation rate decreases with an increased mass flow rate (Arefin et al., 2020). The water loss due to evaporation with the variation of solar intensity and heat of panel is shown in Fig. 20.

The water losses in a sheet and tube type PVT systems are almost negligible as it uses the closed-loop structure. The factors affecting evaporation rate in PV panel cooling using wick structure depend on the arrangement of wick structure on the backside of PV panel, solar intensity and temperature of PV panel.

7. Economic analysis of photovoltaic thermal system

Photovoltaic thermal system performance depends upon the

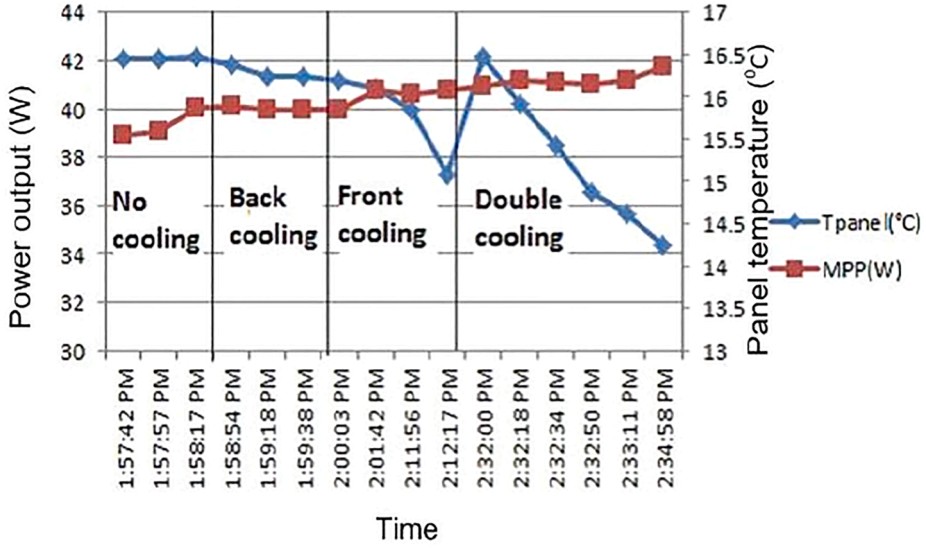


Fig. 19. Effect of the front, back and dual surface cooling on the panel temperature and power output (Hachicha et al., 2015).

Table 3
Literature summary for front and back surfaces cooling of PV panel.

Authors	Characteristics of PV panel	Cooling approach	Mass flow rates	Electrical Efficiency	Thermal Efficiency	Efficiency gain	Input water temperature	Energy used for operating the cooling system
(Nižetić et al., 2016)	Type: Monocrystalline Power: 50 W Area: 0.31 m^2	Active	144–225 lit/hr	15.92%	–	2.5% (front surface cooling) 3.6% (back surface cooling) 5.9% (simultaneous cooling)	17 °C	~8.1–12.6 Wh
(Ahmad et al., 2018)	Type: Monocrystalline Power: 80 W Area: 0.643 m^2	Active	20 lit/min	10.35%	–	10%	24 °C	~90Wh
(Hachicha et al., 2015)	Type: Polycrystalline Power: 20 W Area: 0.173 m^2	Passive	–	10.99%	–	3.6% (front surface cooling) 4% (dual surface cooling)	–	Nil

evaluation of thermal and electrical efficiencies and economic performances measured in terms of life cycle cost (LCC) and simple payback period (SPBP) and environmental welfares indicated by an Energy Payback period (EPBP) and Greenhouse Payback period (GPBP). Buonomano et al. (2016) investigated the yearly, weekly and daily economic analysis of photovoltaic thermal collectors. The PVT system's generated electrical and thermal energy can save about 650 €/year. Also, a simple payback period (SPBP) of 8 and 7 years is required for PVT and PV systems. Taking into account capital cost contribution, SPBP was adjusted to 4 years. Chow and Ji (2012) presented a life cycle analysis by evaluation of the energy payback period and greenhouse payback period for building integrated (BIPVT) and free standalone PVT systems for the climatic condition of Hong Kong. The assessment shows that the EPBP of the BIPVT was 3.8 years and the free standalone PVT was 2.8 years. Both the system's performance was better than the standalone PV system for BIPV and free standalone PV in terms of GPBP of 4.0 and 3.2 years, respectively. The life cycle assessment of the PVT system in the Indian environmental condition for solar water heater application shows that a simple payback period (SPBP) in Delhi and Bangalore was 10.41 years and 5.20 years for 150 days and 300 days use in a year, respectively. Considering annual energy savings by the PVT system, EPBP of 1.3 years was found, reducing CO_2 emissions by 4.3 tons (Dubey and Tiwari, 2008; Maheshwari and Jain, 2017). The economic analysis for glazed and unglazed PVT systems (electrical and thermal performance) for one year was presented by Somasundaram and Tay (2019). Compared to unglazed PVT(8%), the thermal performance of glazed PVT (26%) is higher with an internal rate of return of 10.2% and 12.5 years of the payback period for full-year energy utilization. The PVT system's annual benefits are $2696 and $4578 for the electrical and solar thermal energy output of 10,235 kWh and 16,511 kWh, respectively. Babu and Ponnambalam (2021) carried out the economical assessment for water type PVT, air type PVT, phase change material (PCM) based PVT, thermoelectric generator (TEG) integrated PVT and other hybrid systems. The study shows that the TEG based PVT system bid a low Levelized cost of energy (LCOE) of 0.287 $/kWh with a panel cost of 0.36 $/W at a low installation cost of 1.18 $/W compared to all other PVT systems. Compared to the standalone PV system, the LCOE of all PVT systems increases from 8.7 to 90% and the system's energy gain is 10 to 50% higher. The TEG integrated PVT system has more energy and economic feasibility for rooftop power generation compared to all other systems. Wang et al. (2019) carried out an economic assessment of a PVT Solar-CHP system, a PVT solar power system, an ETC solar thermal system and combined PV-ETC Solar-CHP systems of installation area of 4000 m^2 for heat and power generation for sports centres. The PVT Solar-CHP system performs better than other systems in terms of total energy output with a thermal and electrical energy yield of 51.3% and 82.3% over a year. The PV system is most profitable as the payback period is 9.4 years and the lowest Levelized cost of energy is 0.089 €/kWh, compared to other

Table 4
Summary of different cooling methods used over the surfaces and their effects.

PV surface	Cooling phenomena used	Effects
Front surface	Producing film of water over the surface	• Less evaporation rate of water • Less quantity of water required • PV panel cleaning improves optical efficiency • The surface temperature drops up to 22 °C
	Spray nozzle system	• Higher evaporation rate of water • Quick removal of heat • A large quantity of water required • PV panel cleaning improves optical efficiency • The surface temperature drops up to 27 °C • Higher electrical efficiency obtained
Back surface	Spray nozzle system	• Extract more heat effectively • Quickly temperature drops • The surface temperature drops up to 23 °C • A large quantity of water required • Higher electrical efficiency obtained
	PVT absorber collector system	• Uniform/Non-uniform cooling • Produces both electrical and thermal energy • The surface temperature drops up to 33 °C during peak hour • Heat extraction increase due to the use of copper tubes • Increase in the overall efficiency of the systems
	Evaporative cooling (using cloths, wick structure, wick structure with extended surfaces)	• Uniform cooling • More suitable for standalone PV • The surface temperature drops up to 20 °C • Extended surfaces improve the heat extraction rate
Dual (Front and back) surface	Water immersion	• Uniform cooling • The surface temperature drops up to 8 °C • efficiency gain about 17.8% at 1 cm water depth • Encourage to use in concentrated photovoltaic systems
	Spray nozzle system on both sides	• Uniform/Non-uniform cooling • Require a large quantity of water • Effective cooling during peak sunshine hour • The average surface temperature drops up to 30 °C • The higher electrical efficiency gain
	Spraying (front surface) and direct water contact (back surface) system	• Effectively cool in hot arid regions • The surface temperature drops up to 30 °C • Removal of hot spot generation due to uniform cooling • Spraying system is more suitable for heat extraction • Removal of dust and impurities from the front surface

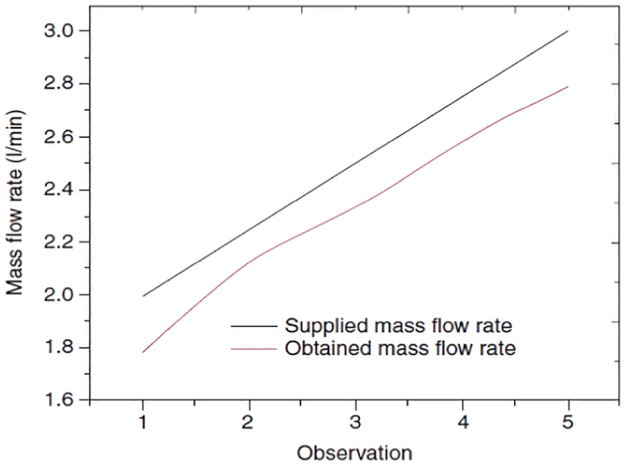


Fig. 20. Losses of water due to evaporation and leakage (Arefin et al., 2020).

systems. On the other hand, the ETC solar thermal system is not economically feasible for the application of sports centres and increasing the share of the ETC area in the combined PV-ETC Solar-CHP system is unfavourable due to the low price of natural gas.

8. Environmental effect of photovoltaic thermal system

To upgrade with new technological revolutions and better practices in the future power system, the PVT system's environmental influences based on solar energy were accessed. There are 32 different influences under the main issues of land, routine and accidental pollution, visibility, effect on buildings, the effect on natural resources, air pollution, sound pollution, waste management, climate and greenhouse gases. Out of the 32 impacts, 22 are favourable to a conventional energy source, 4 impacts are neutral, zero are harmful and 6 need further examination (Tsoutsos et al., 2005; Turney and Fthenakis, 2011). It has been stated that solar PVT power plants installed in a hot arid region having high solar irradiance and no wildlife will be the best solution from an environmental perspective. The cooled photovoltaic system will reduce environmental emission and mitigate the health problem resulting from fossil fuels. The solar power technologies are static and emit 14–45 g of CO_2 per kW/h (Hernandez et al., 2014) and the use of photovoltaic systems can reduce CO_2 by 69–100 million tons, NO_X by 68–99 thousand tons and SO_2 by 126–184 thousand tons by 2030 (Hosenuzzaman et al., 2015).

9. Discussion and scope for research

In this paper, the front surface, back surface and combined front and back surface cooling of the PV system have been discussed. The effects and causes of high working temperature on photovoltaic module performance and various cooling techniques using the active and passive cooling approaches have been described briefly. The cooling technology adopted by various authors enhances the PV module electrical and thermal performance described in detail. Different researchers performed theoretical, numerical and experimental analyses to evaluate water-based PV systems efficiency using different designs of solar

thermal absorbers and various cooling phenomena with active and passive cooling approaches. PVT system performance mostly depends upon inlet water temperature, water flow rate, the geometry shape and water channel size. For 20 °C water inlet temperature, the water outlet temperature rises to 40 °C in a serpentine absorber collector and the PVT system's thermal efficiency gains by sheet and tube type collector. The thermal efficiencies lower in the serpentine absorber collectors compared to sheet and tube type absorber collectors.

After going through the literature, it is observed that the researchers used either the front or back surface of the panel and either active or passive cooling method to cool the surfaces, but not both the surfaces simultaneously. The effect of a combined front and back surfaces cooling with different cooling phenomena on photovoltaic performance has not been explored so far. Some researchers studied the effect of PV panel cooling from both sides in the hot arid region. Extensive research is required to develop cooling techniques to sustain and cool the PV module in higher solar insolation regions. The PV/PCM-PVT system could be the optimal solution for maintaining the PV panel's temperature in the hot arid region. PVT system is best suited to improve the PV electrical and thermal performance. This system is suitable for integration with building integrated photovoltaic. The PVT system is the most appropriate technology for enhancing the life cycle cost, simple payback period and energy payback period. This increases the PV module's life and efficiency and meets the hot water requirement for household and industrial purposes.

10. Conclusion

The effect of the front, back and combined front and back surface cooling on the performance of the PV system using water is discussed in this paper. This paper also discusses the different designs of solar thermal absorber collectors, technical aspects, cooling methods and development in the cooling system of PV panels. This study was conducted to highlights the importance of development in photovoltaic thermal systems to enhance electrical and thermal efficiency and the following conclusions were drawn.

- Continuous water flow over the top surface effectively cools the PV panel up to 22 °C and removes dust on the panel. Thus, it improves optical efficiency.
- For improving the efficiency of photovoltaic panels and generating hot water, a solar thermal absorber collector system is best the solution.
- The combined cooling of the front and back surface with active and passive cooling approach significantly reduces the panel's temperature up to 30 °C in a hot arid region.
- PV/PCM-PVT water-based system with the active and passive cooling approach is a more appropriate solution in the high solar insolation region, but needs further investigation. Also, it is an effective solution for increasing the demand of storing heat and hot water generation.
- The effect of solar irradiance, ambient temperature, wind speed and position plays a significant role in the PV module cooling.
- A Hybrid PV cooling system is more effective in CO_2 emission reduction to the atmosphere than the PV system without cooling.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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